

SMART TO-DO LIST APPLICATION USING PYTHON

Internship Mini Project Report

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Internship: Machine Learning Internship at Geonixa

Technology Used: Python

1. Introduction

The Smart To-Do List Application is a Python-based mini project developed to manage daily tasks efficiently.

This application allows users to:

- Add Tasks
- View Tasks
- Update Tasks
- Delete Tasks
- Search Tasks
- Mark Tasks as Completed
- Manage Priority Levels

The project also uses file handling to store tasks permanently.

This project was developed as part of the internship program to improve practical programming and problem-solving skills.

2. Objectives

- To understand Python programming concepts
- To implement file handling operations
- To use functions and loops effectively
- To apply exception handling
- To develop a real-time mini application

3. Features of the Project

- Add Tasks
 - View Tasks
 - Update Tasks
 - Delete Tasks
 - Search Tasks
 - Mark Tasks as Completed
 - Priority Levels for Tasks
 - Save Tasks using File Handling
 - Clear All Tasks
-

4. Technologies Used

Technology	Purpose
Python	Main Programming Language
File Handling	Store tasks permanently
Functions	Modular Programming
Exception Handling	Error Management

5. Project Explanation

The project stores all tasks in a text file named: t1.txt

Users can perform different operations through a menu-driven interface. The application provides:

- Task Management
- Task Searching
- Task Completion Tracking
- Priority Management

The tasks remain saved even after closing the application because of file handling.

6. Sample Output

===== SMART TO-DO LIST =====

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Complete Task
6. Search Task
7. Clear All Tasks
8. Exit

7. Sample Tasks

1. [Pending][High] Wake Up early at morning
 2. [Completed][Medium] Complete Python Mini Project
 3. [Pending][Low] Update LinkedIn Profile
-

8. Python Source Code

```
import os

TASKS_FILE = "t1.txt"

def load_tasks():
    if not os.path.exists(TASKS_FILE):
        return []

    with open(TASKS_FILE, 'r') as file:
        return [task.strip() for task in file.readlines()]

def save_tasks(tasks):
    with open(TASKS_FILE, 'w') as file:
        for task in tasks:
            file.write(task + "\n")

def add_task():
    task = input("Enter task: ")
    priority = input("Enter priority (High/Medium/Low): ")

    tasks = load_tasks()
    tasks.append(f"[Pending] [{priority}] {task}")

    save_tasks(tasks)
    print("Task added successfully")

def view_tasks():
    tasks = load_tasks()

    if not tasks:
        print("No tasks found")
    else:
        print("\n===== YOUR TO-DO LIST =====")

        for i, task in enumerate(tasks, 1):
            print(i, ".", task)

        print("\nTotal Tasks:", len(tasks))

def update_task():
    view_tasks()
    tasks = load_tasks()

    try:
        index = int(input("Enter task number to update: ")) - 1

        if 0 <= index < len(tasks):
            new_task = input("Enter new task: ")
            priority = input("Enter priority: ")

            tasks[index] = f"[Pending] [{priority}] {new_task}"

            save_tasks(tasks)
            print("Task updated successfully")
        else:
            print("Enter valid task number")

    except ValueError:
        print("Enter numbers only")

def delete_task():
    view_tasks()
    tasks = load_tasks()

    try:
        index = int(input("Enter task number to delete: ")) - 1

        if 0 <= index < len(tasks):
            removed = tasks.pop(index)
```

```

        save_tasks(tasks)
        print("Deleted:", removed)
    else:
        print("Invalid task number")

except ValueError:
    print("Enter numbers only")

def complete_task():
    view_tasks()
    tasks = load_tasks()

    try:
        index = int(input("Enter completed task number: ")) - 1

        if 0 <= index < len(tasks):
            tasks[index] = tasks[index].replace("[Pending]", "[Completed]")

            save_tasks(tasks)
            print("Task marked as completed")
        else:
            print("Invalid task number")

    except ValueError:
        print("Enter numbers only")

def search_task():
    tasks = load_tasks()

    keyword = input("Enter keyword to search: ").lower()
    found = False

    for task in tasks:
        if keyword in task.lower():
            print(task)
            found = True

    if not found:
        print("No matching tasks found")

def clear_tasks():
    confirm = input("Are you sure? (yes/no): ")

    if confirm.lower() == "yes":
        save_tasks([])
        print("All tasks cleared")
    else:
        print("Operation cancelled")

while True:
    print("\n===== SMART TO-DO LIST =====")

    print("1. Add Task")
    print("2. View Tasks")
    print("3. Update Task")
    print("4. Delete Task")
    print("5. Complete Task")
    print("6. Search Task")
    print("7. Clear All Tasks")
    print("8. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        add_task()

    elif choice == "2":
        view_tasks()

    elif choice == "3":
        update_task()

    elif choice == "4":
        delete_task()

    elif choice == "5":
        complete_task()

```

```

elif choice == "6":
    search_task()
elif choice == "7":
    clear_tasks()

elif choice == "8":
    print("Thank you for using Smart To-Do List")
    break

else:
    print("Invalid choice")

```

9. Output Screenshots

```

===== SMART TO-DO LIST =====

```

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Complete Task
6. Search Task
7. Clear All Tasks
8. Exit

Enter your choice: 1

Enter task: Wake Up early at morning

Enter priority (High/Medium/Low): High

Task added successfully

```

===== SMART TO-DO LIST =====

```

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Complete Task
6. Search Task
7. Clear All Tasks
8. Exit

Enter your choice: 1

Enter task: Complete Python Mini Project

Enter priority (High/Medium/Low): High

Task added successfully

```

===== SMART TO-DO LIST =====

```

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Complete Task
6. Search Task
7. Clear All Tasks
8. Exit

Enter your choice: 2

```

===== YOUR TO-DO LIST =====

```

- 1 . [Pending] [High] Wake Up early at morning
- 2 . [Pending] [High] Complete Python Mini Project
- 3 . [Pending] [Medium] Practice DSA Problems
- 4 . [Pending] [Low] Update LinkedIn Profile
- 5 . [Pending] [Low] Watch Python Tutorial

Total Tasks: 5

===== SMART TO-DO LIST =====

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Complete Task
6. Search Task
7. Clear All Tasks
8. Exit

Enter your choice: 3

===== YOUR TO-DO LIST =====

- 1 . [Pending] [High] Wake Up early at morning
- 2 . [Pending] [High] Complete Python Mini Project
- 3 . [Pending] [Medium] Practice DSA Problems
- 4 . [Pending] [Low] Update LinkedIn Profile
- 5 . [Pending] [Low] Watch Python Tutorial

Total Tasks: 5

Enter task number to update: 5

Enter new task: Prepare Revision Notes

Enter priority: High

Task updated successfully

===== SMART TO-DO LIST =====

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Complete Task
6. Search Task
7. Clear All Tasks
8. Exit

Enter your choice: 2

===== YOUR TO-DO LIST =====

- 1 . [Pending] [High] Wake Up early at morning
- 2 . [Pending] [High] Complete Python Mini Project
- 3 . [Pending] [Medium] Practice DSA Problems
- 4 . [Pending] [Low] Update LinkedIn Profile
- 5 . [Pending] [High] Prepare Revision Notes

Total Tasks: 5

===== SMART TO-DO LIST =====

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Complete Task
6. Search Task
7. Clear All Tasks
8. Exit

Enter your choice: 4

===== YOUR TO-DO LIST =====

- 1 . [Pending] [High] Wake Up early at morning
- 2 . [Pending] [High] Complete Python Mini Project
- 3 . [Pending] [Medium] Practice DSA Problems
- 4 . [Pending] [Low] Update LinkedIn Profile
- 5 . [Pending] [High] Prepare Revision Notes

Total Tasks: 5

Enter task number to delete: 4

Deleted: [Pending] [Low] Update LinkedIn Profile

===== SMART TO-DO LIST =====

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Complete Task
6. Search Task
7. Clear All Tasks
8. Exit

Enter your choice: 2

===== YOUR TO-DO LIST =====

- 1 . [Pending] [High] Wake Up early at morning
- 2 . [Pending] [High] Complete Python Mini Project
- 3 . [Pending] [Medium] Practice DSA Problems
- 4 . [Pending] [High] Prepare Revision Notes

Total Tasks: 4

===== SMART TO-DO LIST =====

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Complete Task
6. Search Task
7. Clear All Tasks
8. Exit

Enter your choice: 5

===== YOUR TO-DO LIST =====

- 1 . [Pending] [High] Wake Up early at morning
- 2 . [Pending] [High] Complete Python Mini Project
- 3 . [Pending] [Medium] Practice DSA Problems
- 4 . [Pending] [High] Prepare Revision Notes

Total Tasks: 4

Enter completed task number: 1

Task marked as completed

===== SMART TO-DO LIST =====

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Complete Task
6. Search Task
7. Clear All Tasks
8. Exit

Enter your choice: 2

===== YOUR TO-DO LIST =====

- 1 . [Completed] [High] Wake Up early at morning
- 2 . [Pending] [High] Complete Python Mini Project
- 3 . [Pending] [Medium] Practice DSA Problems
- 4 . [Pending] [High] Prepare Revision Notes

Total Tasks: 4


```
===== SMART TO-DO LIST =====
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Complete Task
6. Search Task
7. Clear All Tasks
8. Exit
Enter your choice: 6
Enter keyword to search: Practice
[Pending] [Medium] Practice DSA Problems
```

```
===== SMART TO-DO LIST =====
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Complete Task
6. Search Task
7. Clear All Tasks
8. Exit
Enter your choice: 7
Are you sure? (yes/no): yes
All tasks cleared
```

```
===== SMART TO-DO LIST =====
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Complete Task
6. Search Task
7. Clear All Tasks
8. Exit
Enter your choice: 2
No tasks found

===== SMART TO-DO LIST =====
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Complete Task
6. Search Task
7. Clear All Tasks
8. Exit
Enter your choice: 8
Thank you for using Smart To-Do List
```

10. Learning Outcomes

Through this project, the following concepts were learned:

- Python Programming
- File Handling
- Functions and Loops
- Exception Handling
- Problem-Solving Techniques
- Real-Time Application Development

11. Conclusion

The Smart To-Do List Application was successfully developed using Python.

The project helped improve practical programming skills and understanding of task management systems.

It also provided hands-on experience with:

- File Handling
- User Interaction
- Real-Time Mini Application Development

12. References

- Python Official Documentation
- Internship Learning Materials
- VS Code Documentation